



INTRODUCTION TO PRODUCTION AND MANUFACTURING SYSTEMS

INDU 211

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Agenda

- Chapter 14 Deterministic Operations Research

Operations Research

- Scientific research on operations
- Started during World War II
- Allocation of limited resources to various operations (activities)
- How to conduct and coordinate operations (activities)?
- The use of quantitative techniques in the domain of decision making
 - Decision support tool
 - Building a mathematical model to abstract the essence of the real problem
 - Solving the model to obtain optimal decisions
- Deterministic vs Probabilistic
- Linear programming
- Discrete optimization (Integer, Binary, and Mixed-integer programming)
- Nonlinear programming

Operations Research

- Applications:
 - Supply chain planning
 - Production planning
 - Inventory planning
 - Facility location
 - Facility layout design
 - Transportation and logistics
 - Personnel scheduling
 - Timetabling
 - Job scheduling
 - Energy management
 - Robotics (job allocation and planning)
 - Product design

Linear Programming (LP)

- Elements of a mathematical programming model:
 - Decision variables
 - Parameters
 - Objective function
 - Constraints
- LP: A mathematical programming model where the objective function and all constraints are linear functions of decision variables
- Linearity is a requirement of the model in both the objective function and constraints

Objective Function

Maximize (or Minimize) $Z = C_1X_1 + C_2X_2 + \dots + C_nX_n$

- C_j is a constant that describes the rate of contribution to costs or profit (of units being produced (X_j))
- X_j Decision variables
- Z is the total cost or profit (from the given number of units being produced)

Constraints

$$A_{11}X_1 + A_{12}X_2 + \dots + A_{1n}X_n \leq B_1$$

$$A_{21}X_1 + A_{22}X_2 + \dots + A_{2n}X_n \geq B_2$$

⋮

$$A_{M1}X_1 + A_{M2}X_2 + \dots + A_{Mn}X_n = B_M$$

- A_{ij} are resource requirements (technological requirements) for each of the related (X_j) decision variables (activities).
- B_i are the available resource requirements.
- The direction of the inequalities can be all or a combination of $\leq$, $\geq$, or $=$ linear mathematical expressions.

Non-Negativity Requirement

$$X_1, X_2, \dots, X_n \geq 0$$

- All linear programming model formulations require their decision variables to be non-negative.

An example of a product mix problem

Lawn Grow Manufacturing Company must determine the mix of its commercial riding mower products to be produced next year. The company produces two product lines, the Max and the Multimax. The average profit is \$400 for each Max and \$800 for each Multimax. Fabrication and assembly are limited resources. There is a maximum of 5,000 hours of fabrication capacity available per month (Each Max requires 3 hours and each Multimax requires 5 hours). There is a maximum of 3,000 hours of assembly capacity available per month (Each Max requires 1 hour and each Multimax requires 4 hours).

Question: How many of each riding mower should be produced each month in order to maximize profit?



The Objective Function

If we define the Max and Multimax products as the two decision variables X_1 and X_2 , and since we want to maximize profit, we can state the objective function as follows:

$$\text{Maximize } Z = 400X_1 + 800X_2$$

Where

Z = the monthly profit from Max and Multimax

X_1 = the number of Max produced each month

X_2 = the number of Multimax produced each month

Constraints

Given the resource information below from the problem:

Max (X1)	Multimax (X2)		
Required Time/Unit	Required Time/Unit	Available Time/Month	
3	5	5,000	Fab
1	4	3,000	Assy

We can now state the constraints and non-negativity requirement:

$$3X_1 + 5X_2 \leq 5,000 \quad (\text{Fab.})$$

$$X_1 + 4X_2 \leq 3,000 \quad (\text{Assy.})$$

$$X_1, X_2 \geq 0 \quad (\text{Non-negativity})$$

Note that the inequalities are less-than-or-equal since the time resources represent the total available resources for production.

Graphical Solution Procedure for two-variable problems

- Set up graph with X_1 on the x-axis and X_2 on the y-axis
- Graph each constraint
- Find optimal point
 - Determine feasible region
 - Choose an arbitrary objective function value
 - Graph objective function
 - Move objective function line parallel to itself until you find the optimal corner point

Complete LP model for lawn tractor production planning

$$\text{Maximize } Z = 400X_1 + 800X_2$$

Subject to

$$3 X_1 + 5 X_2 \leq 5000$$

$$X_1 + 4 X_2 \leq 3000$$

$$X_1, X_2 \geq 0$$

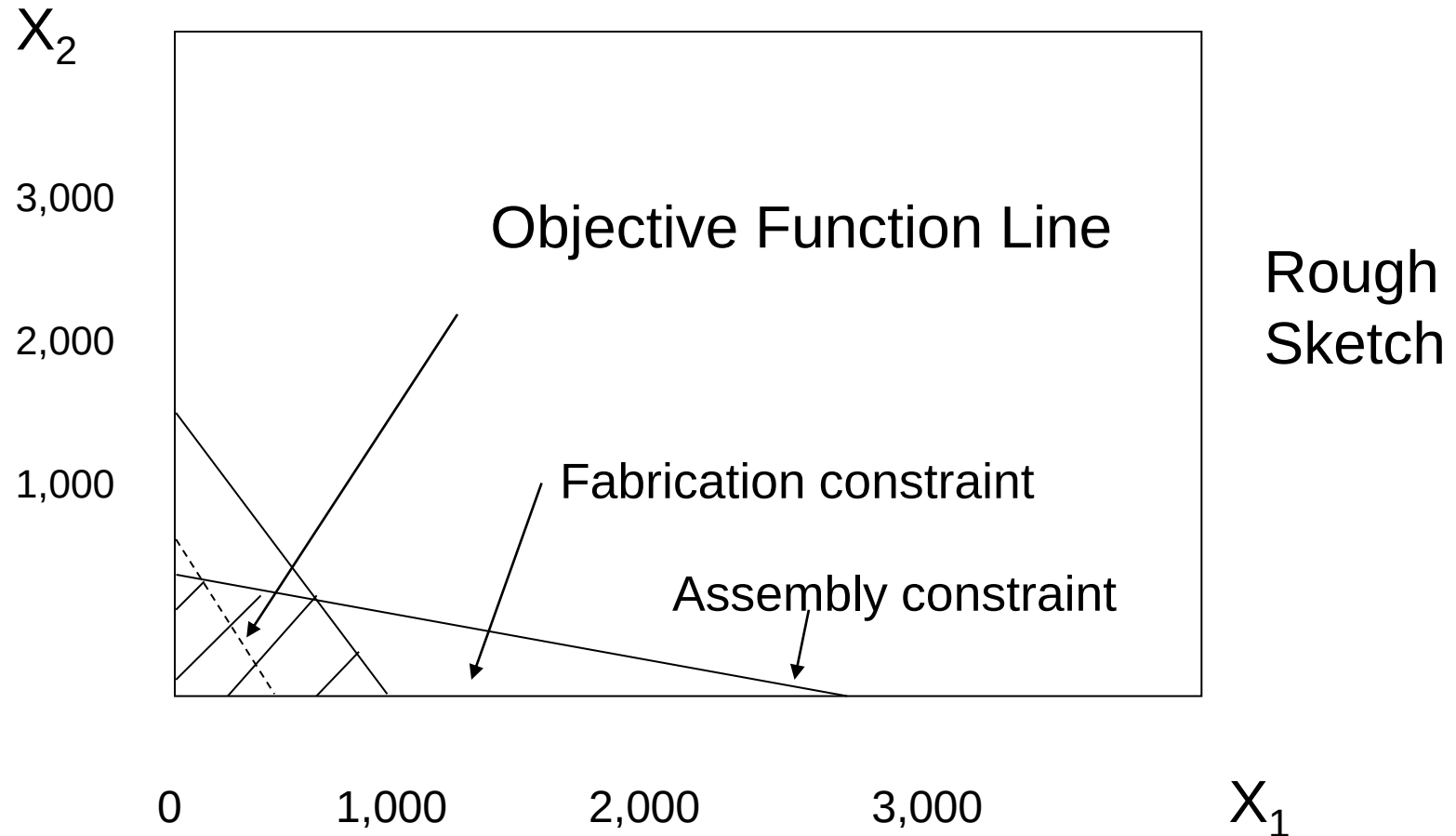
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Graphical Solution Procedure for two-variable problems

1. Let $3X_1 + 5X_2 = 5000$ to find out where it intersects with X_1 and X_2 axes, respectively
2. Let $X_1 + 4X_2 = 3000$ to find out where it intersects with X_1 and X_2 axes, respectively
3. Draw them in a graph, the common area makes “ $\leq$ ” to happen for both lines is the feasible region
4. Let $400X_1 + 800X_2 = \text{any number} > 0$, say 8,000 to find out the direction it moves to increase the most
5. Move the line from Step 4 towards the direction of increasing profit until it hits the border point of the feasible region
6. That “corner point” is the optimal solution of the LP model

Graphical Method



Solution

The math solution can be found by solving the group of the two constraint equations:

$$3X_1 + 5X_2 = 5000$$

$$X_1 + 4X_2 = 3000$$

The solution is: $X_1 = 715$, $X_2 = 571$

It tells that Lawn Grow Manufacturing Company should produce 715 Max and 571 Multimax per month for a profit of

$$Z = 400 \times X_1 + 800 \times X_2 = 400 \times 715 + 800 \times 571 = 742,800 \text{ (dollars)}$$

Example 2- Minimization problem

$$\text{Min } Z = 5x_1 + 4x_2$$

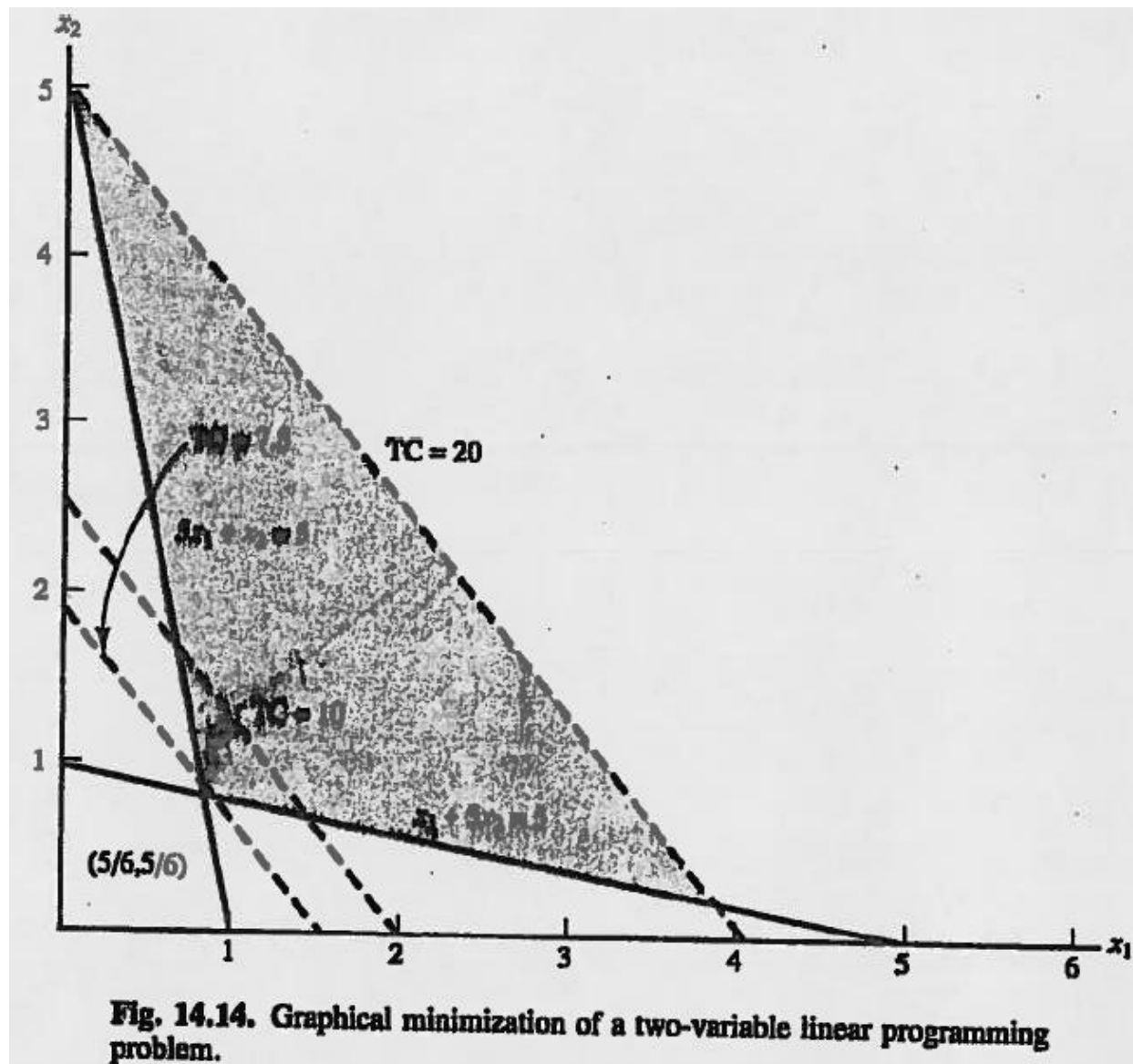
Subject to

$$5x_1 + x_2 \geq 5,$$

$$x_1 + 5x_2 \geq 5,$$

$$x_1, x_2 \geq 0.$$

Graphical method



Typical deterministic OR Models

- Non-linear programming (NLP)
- Integer and Mixed-Integer programming (IP/MIP)
- Linear programming problems (LP)
 - Transportation problems
 - Assignment problems
 - Shortest path problems
 - Minimum spanning tree problems
- TSP/VRP problems
- Dynamic programming (DP)

Problem solving using OR tools

- Problem conceptualization
- Modeling
 - Data, functions, simplifications
- Model verification and validations
- Design/develop a solution method
 - Optimization, heuristic, etc
- Code the model/heuristic using CPLEX, Gurobi, Matlab, Python, etc.
- Test/validate the solution method
- Solve the model to obtain feasible/good/optimal solutions
- Use the model solution to solve the problem